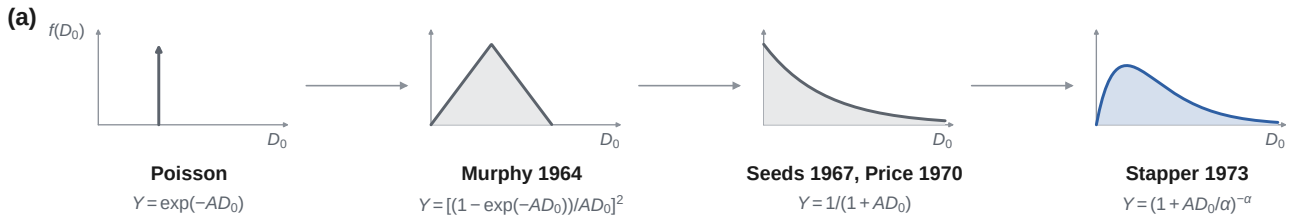
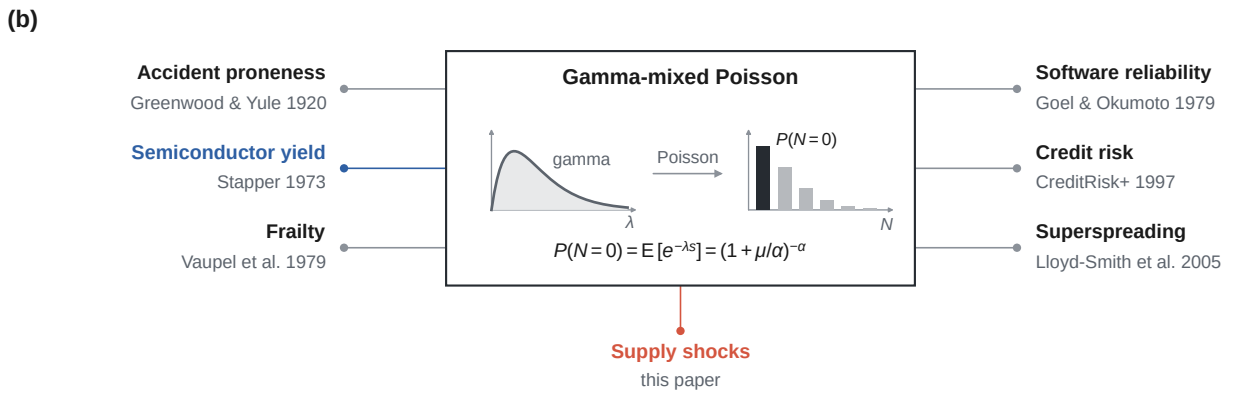

(a)
f(D0)
D0
Poisson
Y = exp(−AD0)
Murphy 1964
Y = [(1 − exp(−AD0))/AD0]2
Seeds 1967, Price 1970
Y = 1/(1 + AD0)
Stapper 1973
Y = (1 + AD0/α)−α


(b)
Accident proneness
Greenwood & Yule 1920
Semiconductor yield
Stapper 1973
Frailty
Vaupel et al. 1979
Gamma-mixed Poisson
gamma
Poisson
P(N = 0)
λ
N
P(N = 0) = E [e−λs] = (1 + μ/α)−α
Software reliability
Goel & Okumoto 1979
Credit risk
CreditRisk+ 1997
Superspreading
Lloyd-Smith et al. 2005
Supply shocks
this paper